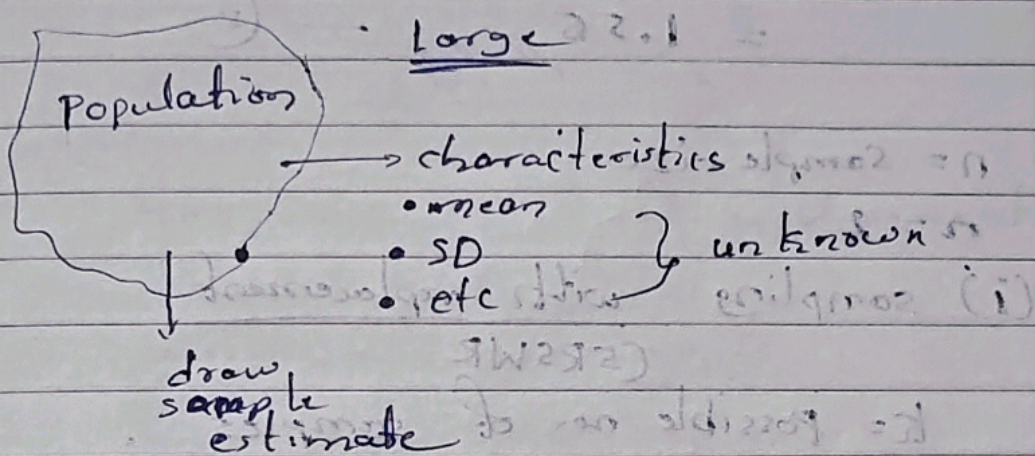
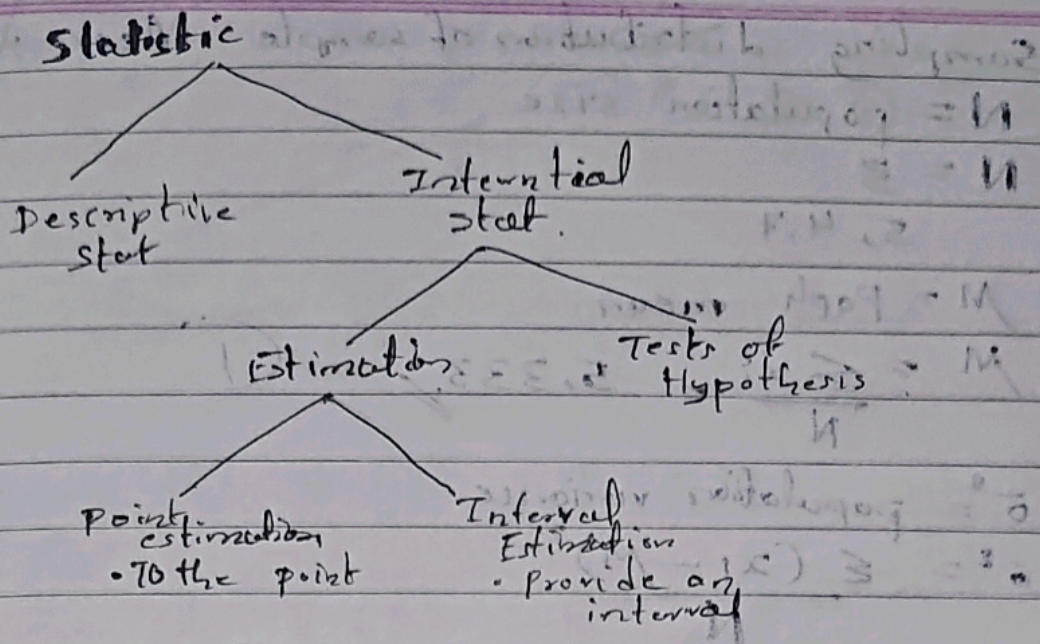




Sampling methods :-

- Simple Random sample (SRS)
- stratified Random sample
- Systematic sample (N, n), (2, n)
- cluster sampling. (N, F), (2, F)

With Replacement (WR) ~~to sample~~ ^{scheme}

Without Replacement (WOR)

Sampling distribution.

Sampling distribution of sample means:

N = population size

$N = 5$

5, 4, 7

M = Popl mean

$$M = \frac{\sum x_i}{N} = \frac{5.333}{5} \quad (1)$$

σ^2 = population variance

$$\sigma^2 = \frac{\sum (x_i - M)^2}{N}$$

$$= \frac{1.56}{5} = 0.312 \quad (2)$$

n = Sample size

$n = 2$

(i) sampling with replacement (SRSWR)

k = Possible no. of samples

$$k = N^n = 3^2 = 9$$

The samples are (5,5), (5,4), (5,7), (4,5), (4,4), (4,7), (7,5), (7,4), (7,7)

(5,5) and (5,4) are not possible as (5+5) $\rightarrow$ 10 > 10

(4,5), (4,4), (4,7), (7,5), (7,4), (7,7) are possible

Means of samples

5, 4.5, 6

4.5, 4, 5.5

6, 5.5, 7

Mean of sample means

$$M_{\bar{x}} = \bar{x} = \frac{\sum \bar{x}_j}{k} = \frac{5.33 \dots (3)}{3}$$

variance of sample means

$$\sigma_{\bar{x}}^2 = \frac{\sum (\bar{x}_j - \bar{x})^2}{k} = \frac{0.78}{3} = 0.26 \quad (4)$$

$$\frac{\sigma^2}{n} = \frac{1.56}{2} = 0.78 \quad (5)$$

From (1) & (3) $\sigma_{\bar{x}} = \sigma_{\bar{x}}$

From (4) & (5) $\sigma^2 = \frac{\sigma^2}{n}$

Case (ii)

sampling without replacement (SRSWOR)

k = possible no. of sample

$$k = \binom{N}{n} = \binom{3}{2} = \frac{3!}{2!1!} = 3$$

The samples are

(5,4), (5,7), (4,7) $\rightarrow$ 7 possible

Means of samples

4.5, 6, 5.5 $\rightarrow$ $\bar{x}_j, j=1, 2, \dots, k$

Means of sample means (2-1) 001

$$M_{\bar{x}} = \bar{x} = \frac{\sum \bar{x}_j}{k} = \frac{5.33}{3} \quad (6)$$

Variance of the sample means

$$\sigma_{\bar{x}}^2 = \frac{\sum (\bar{x}_j - \bar{x})^2}{k} = \frac{0.39}{3} \quad (7)$$

$$\frac{N-n}{N-1} \cdot \frac{\sigma^2}{n} = \frac{3-2}{3-1} \cdot \frac{1.56}{2} = 0.39 \quad (8)$$

From (1) $\bar{X} = \mu$
 From (2) $\bar{X} = \mu + \frac{\sigma^2}{N-1}$

$\sigma^2 = \frac{N-1}{N-1} \cdot \frac{\sigma^2}{N-1}$

Interval estimation (Confidence)
 $\theta \rightarrow$ population characteristic (unknown)

sample (n)
 $\hat{\theta} \rightarrow$ statistic.

$\theta = \hat{\theta} \pm \text{error}$
 $\hat{\theta} - \text{error} \leq \theta \leq \hat{\theta} + \text{error}$

$\theta \in [\hat{\theta} - \text{error}, \hat{\theta} + \text{error}]$

confidence

Probability $P[1 \leq \theta \leq U] = 1 - \alpha$ confidence level

Parameter := statistic $\pm$ error
 $\mu \rightarrow$ population mean

100 (1- α) % confidence interval
 for population mean μ

$\bar{x} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$
 sample mean $\bar{x}$ $\frac{\sigma}{\sqrt{n}}$ population std $Z_{\alpha/2}$ critical value at α level

if population variance is unknown
 (σ^2)

$\hat{\sigma}^2 = S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$ unbiased estimator of σ^2

S.E. ($\bar{x}$) = $\sqrt{\frac{S^2}{n}} = \frac{S}{\sqrt{n}}$

100 (1- α) % C.I for population mean μ
 $\bar{x} - t_{n-1, \alpha/2} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{n-1, \alpha/2} \cdot \frac{S}{\sqrt{n}}$

t -distribution critical values at α level with $n-1$ degrees of freedom

d - actual distance between star & observations

$\bar{d} = 2$ light year

error = ± 0.5

$X \rightarrow$ measurement of distance

$x_1, x_2, x_3, \dots, x_n$

$\bar{x} = \frac{\sum x_i}{n} \rightarrow$ estimate of d

confidence = $1 - \alpha \geq 95\%$

$n = ?$

error = $\bar{x} - d$

$-0.5 \leq \text{error} \leq 0.5$

i.e. $-0.5 \leq \bar{x} - d \leq 0.5$

i.e. $P(-0.5 \leq \bar{x} - d \leq 0.5) \geq 0.95$

From sampling dist of $\bar{x}$

Hypothesis (H) :-

it is assertion abt population parameters

Null Hypothesis (H_0)

$$H_0 : \mu = \mu_0 \quad \text{or} \quad \mu - \mu_0 = 0$$

$$H_0 : \mu_1 = \mu_2 \quad \text{or} \quad \mu - \mu_2 = 0$$

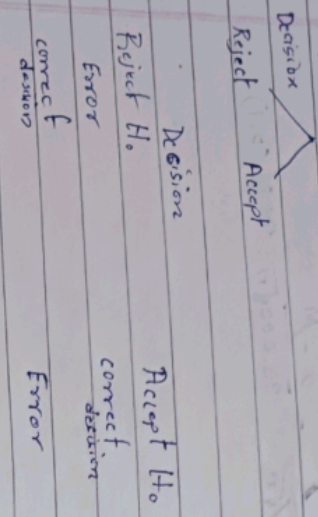
Test H_0

Alternative Hypo (H_1)

$$H_0 : \mu = \mu_0 \rightarrow H_1 : \mu \neq \mu_0$$

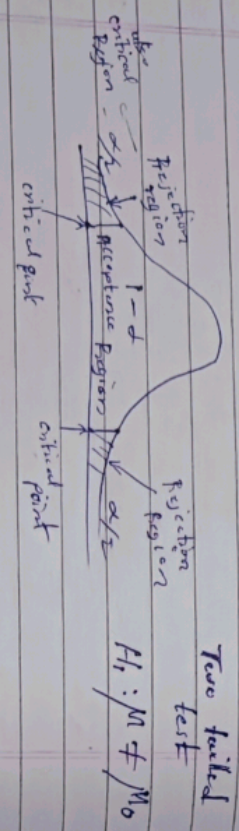
$$H_1 : \mu_1 \neq \mu_2$$

Test H_0 vs H_1



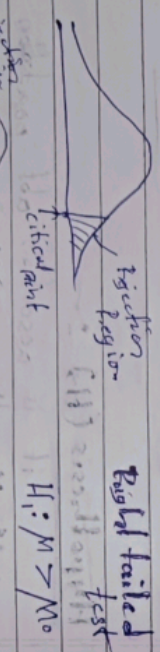
Type I error = $P(\text{Reject } H_0 / H_0 \text{ is true}) = \alpha$
 Type II error = $P(\text{Accept } H_0 / H_1 \text{ is true}) = \beta$

Type II error = $P(\text{Accept } H_0 / H_1 \text{ true})$
 $P(\text{Type - II error}) = P(\text{Accept } H_0 | H_1 \text{ true}) = \beta$



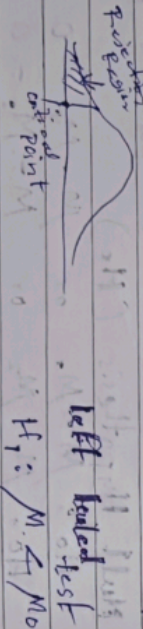
Two tailed test

$H_1: \mu \neq \mu_0$



Right tailed test

$H_1: \mu > \mu_0$



Left tailed test

$H_1: \mu < \mu_0$

Test for single mean

one population one sample

Population $N(\mu, \sigma^2)$

$H_0: \mu = \mu_0$ (unknown)

vs

$H_1: \mu \neq \mu_0$ (Two tailed test)

draw a sample of size n from population
 $\bar{X} = \bar{x} \sim N(\mu, \frac{\sigma^2}{n})$

Case 1: Population variance is known

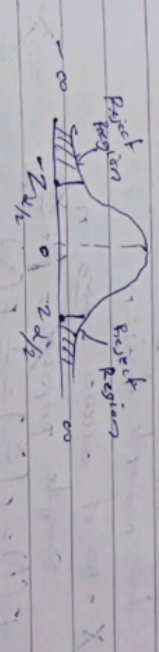
i.e. σ^2 is known

Test stat: under H_0

$$Z_0 = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$

let α be the level of significance

Critical value = $\pm z_{\alpha/2}$



Decision rule: if $Z_0 < -z_{\alpha/2}$ or $Z_0 > z_{\alpha/2}$, then Reject H_0

at α level (1%) otherwise we fail to reject H_0

100 (1-alpha)% C.I for pop'n mean μ

known σ^2

$$\bar{x} - \frac{\sigma}{\sqrt{n}} \cdot z_{\alpha/2} \leq \mu \leq \bar{x} + \frac{\sigma}{\sqrt{n}} \cdot z_{\alpha/2}$$

Unknown σ^2

$$s^2 = S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

α	one tail	Two tail
0.95	1.64	1.96
0.01	2.33	2.58

$$\therefore 95\% \text{ C.I for } \mu_1 - \mu_2 \text{ is}$$

$$(36.2 - 2.9) - (0.1185)(1.96) \leq \mu_1 - \mu_2 \leq (36.2 - 2.9) + (0.1185)(1.96)$$

Test for single mean

Case 1 σ^2 is known

Case 2: σ^2 is unknown

$H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$
(Two tailed Test)

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$\text{Test stat: } t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

replace all σ with S in critical value by T distribution critical value in decision rule

3(b)

$$n = 100$$

$$\bar{x} = 37.7$$

$$\sigma = 9.2$$

$$\mu_0 = 35$$

$H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$ (right tailed test)

Test stat:

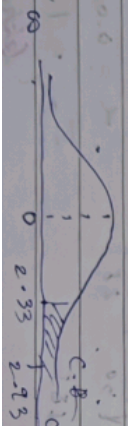
$$Z_0 = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$\frac{37.7 - 35}{9.2/\sqrt{100}}$$

$$= 2.9348$$

$$\alpha = 0.01$$

critical value = $Z_{\alpha} = 2.33$



16 L.P :-

$$\mu_0 = 5000, \sigma = 800$$

$$n = 100, \bar{x} = 5200$$

$H_0: \mu = \mu_0$ vs $H_1: \mu > \mu_0$

Test stat: $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$

$$Z = \frac{5200 - 5000}{800/\sqrt{100}}$$

$$= 2.5$$

$$\alpha = 0.05$$

$$\text{critical value} = Z_{\alpha} = 1.64$$



Reject H_0 at 0.05 level

Conclusion: No. of defective wires has increased

20 L.P

X = breaking strength of copper wires

$$n = 10$$

$\mu_0 = 548$ kg wt. of lot

$\mu_1 = 578$ kg wt. of lot

Observation: Population variance σ^2 is unknown

$H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$ (Two tailed Test)

Test stat: $t_0 = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

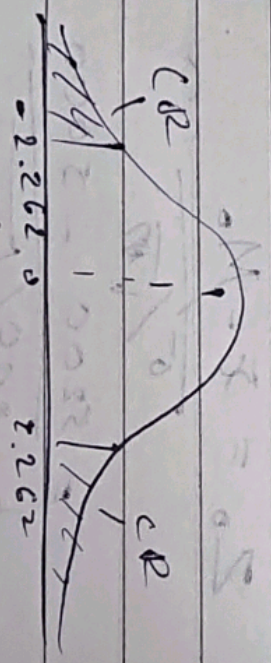
$\bar{x} = \frac{\sum x_i}{n} = \frac{571.7}{10} = 57.17$

$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{134.6778}{9} = 14.9642$

$s = \sqrt{14.9642} = 3.868$

Let $\alpha = 0.05$ Critical value $t_{n-1, \alpha/2} = t_{9, 0.025} = 2.262$

d.f. $n-1 = 10-1 = 9$



$\therefore -2.262 \leq -1.7167 \leq 2.262$

$\therefore$ We fail to reject H_0 at 0.05 level

Conclusion: Mean breaking strength of the lot be considered as 57 kg wt.

LP 19

South

North

82.5% =

82.5% = P_2

$$n_1 = 300$$

$$n_2 = 200$$

$$P_1 = 0.56$$

$$P_2 = 0.48$$

a) $H_0: P_1 = P_2$ vs $H_1: P_1 \neq P_2$ (Two tailed)

Test stat.:

$$Z_0 = \frac{P_1 - P_2}{\sqrt{P_1(1-P_1) + P_2(1-P_2)}}$$

$$\sim N(0,1)$$

Then

we fail

0.5.

White

$$P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

There is no significant difference

$$= \frac{(300 \times 0.56) + (200 \times 0.48)}{300 + 200}$$

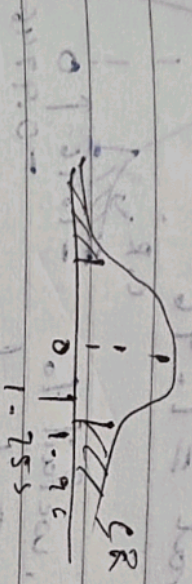
$$= 0.5280$$

$$\therefore Q = 1 - P = 0.4720$$

$$\therefore Z_0 = \frac{0.56 - 0.48}{\sqrt{0.5280 \times 0.4720 \left(\frac{1}{300} + \frac{1}{200} \right)}}$$

$$= 1.755$$

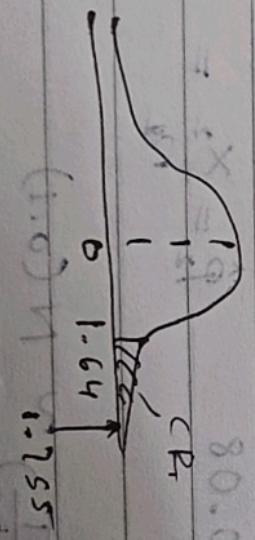
$$\text{let } \alpha = 0.05 \quad Z_{\alpha/2} = 1.96$$



we fail to reject H_0 at 0.05 level

ii) $H_1: P_1 > P_2$ (Right tailed test)

critical = $Z_{\alpha} = 1.64$



$\therefore Z_0 > Z_{\alpha}$ reject H_0 at 0.05 level

Q22

$P_0 = 0.70, n = 15$

$X = \text{no. of homes without head pumps} = 8$

$\therefore \hat{p} = \frac{X}{n} = \frac{8}{15} = 0.533$

$H_0: P = P_0 \text{ vs } H_1: P \neq P_0 \text{ (Two tailed)}$

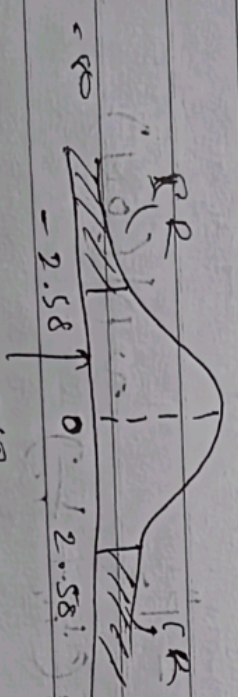
Test stat:

$Z_0 = \frac{\hat{p} - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} \sim N(0,1)$

$= \frac{0.533 - 0.7}{\sqrt{\frac{0.7 \times 0.3}{15}}} = -1.4368$

$\sqrt{\frac{0.7 \times 0.3}{15}}$

$\alpha = 0.01, \text{ critical} = Z_{\alpha/2} = 2.578$



$\therefore Z_{\alpha/2} \leq Z_0 \leq Z_{\alpha/2}$ we fail to reject H_0 at 0.01 level

Conclusion: Bidder is true

2 test Z_0 (Left, right, two)

Test for different of two population proportion

$P_1 \rightarrow$ Proportion in 1st Population

$P_2 \rightarrow$ Proportion in 2nd Population

$$H_0: P_1 = P_2 \quad \text{vs} \quad P_1 \neq P_2$$

$\hat{P}_1 =$ sample proportion 1st population

$\hat{P}_2 =$ sample proportion 2nd population

$$\hat{P}_1 = \frac{X_1}{n_1}$$

$$\hat{P}_2 = \frac{X_2}{n_2}$$

Test stat : under $H_0: P_1 = P_2$

$$Z_0 = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{P_0 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim N(0,1)$$

where $P = \frac{X_1 + X_2}{n_1 + n_2} = \frac{n_1 \hat{P}_1 + n_2 \hat{P}_2}{n_1 + n_2}$

$$Q = 1 - P$$